DBMS – Project

"A Unified Platform for Modern Court Management"

❖ Group Members

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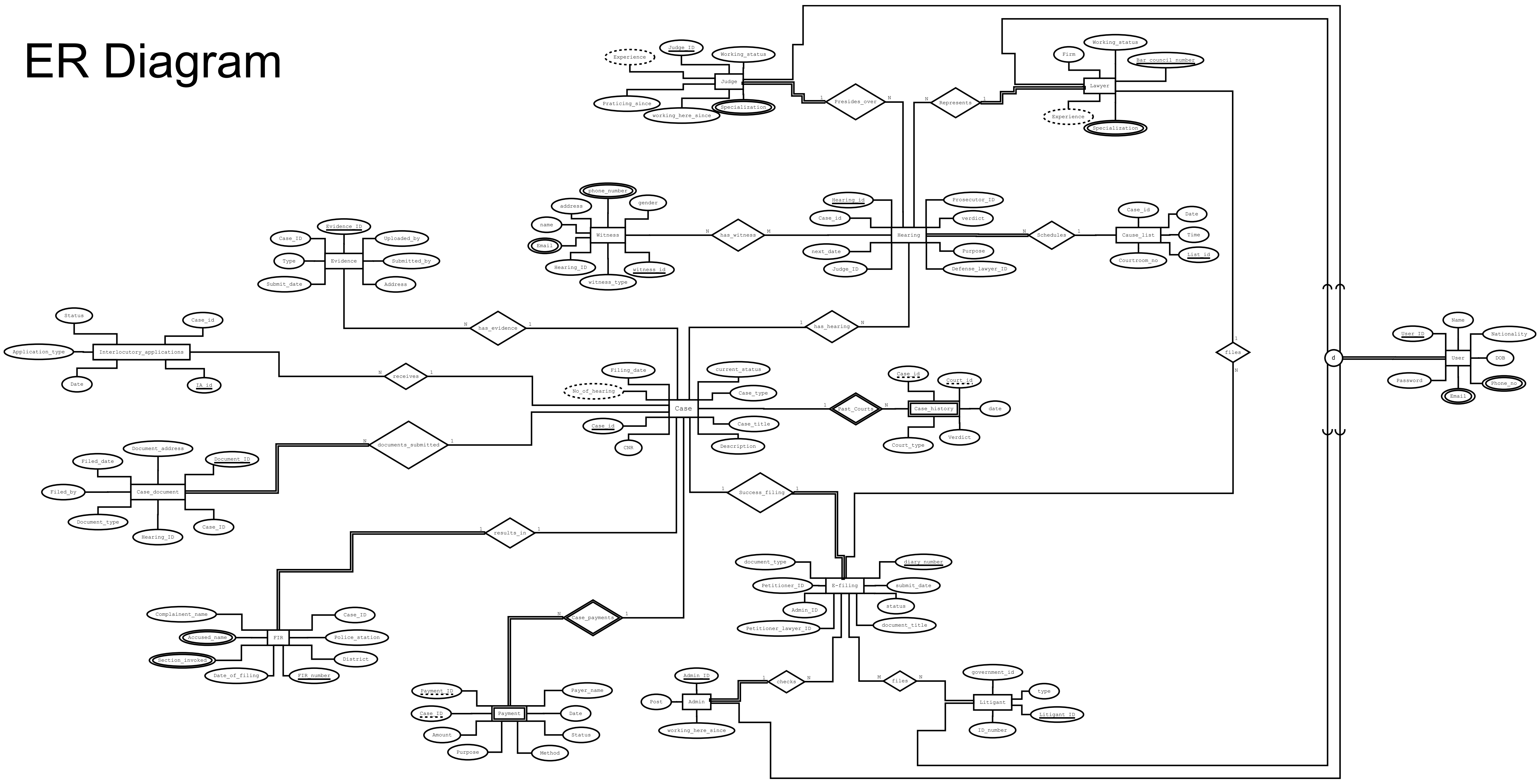
❖ **For any .png, .dia, or pdf files refer to the below link:**
https://github.com/Darpan0610/DBMS_Project/

ER Diagram



Judge
Judge_ID
Experience
Working_status
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working_here_since
Specialization
Presides_over
Represents
Lawyer
Firm
Working_status
Bar_council_number
Experience
Specialization
Witness
phone_number
address
gender
name
Email
Hearing_ID
witness_id
witness_type
has_witness
Hearing
Hearing_id
Case_id
next_date
Judge_ID
Prosecutor_ID
verdict
Purpose
Defense_lawyer_ID
Schedules
Cause_list
Case_id
Date
Time
List_id
Courtroom_no
Evidence
Evidence_ID
Case_ID
Type
Submit_date
Uploaded_by
Submitted_by
Address
has_evidence
has_hearing
files
Interlocutory_applications
Status
Case_id
Application_type
Date
IA_id
receives
Case
Filing_date
No_of_hearing
Case_id
CNR
current_status
Case_type
Case_title
Description
Past_Courts
Case_history
Case_id
Court_id
date
Court_type
Verdict
documents_submitted
Case_document
Document_address
Filed_date
Document_ID
Filed_by
Document_type
Hearing_ID
Case_ID
Success_filing
results_in
FIR
Complainent_name
Accused_name
Section_invoked
Date_of_filing
FIR_number
Case_ID
Police_station
District
Case_payments
Payment
Payment_ID
Case_ID
Amount
Purpose
Method
Payer_name
Date
Status
E-filing
document_type
Petitioner_ID
Admin_ID
Petitioner_lawyer_ID
diary_number
submit_date
status
document_title
Admin
Admin_ID
Post
working_here_since
checks
files
Litigant
government_id
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d
User
User_ID
Name
Nationality
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Password
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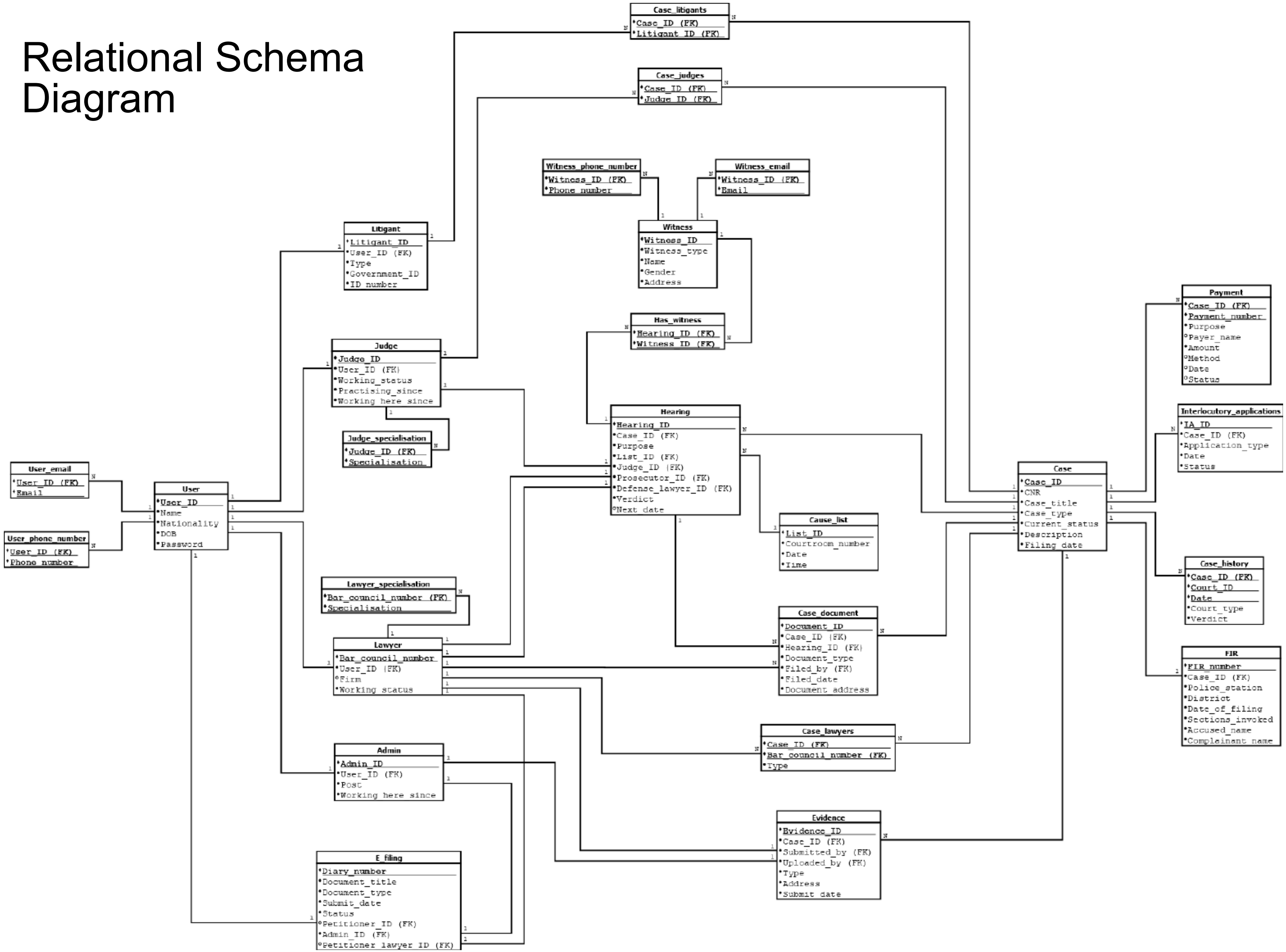
Phone_no

Relational Schema Diagram



➢ BCNF Proofs

1. USER

Attributes: User_ID, Name, Nationality, DOB, Password
Candidate Key(s):
{User_ID}
All FDs:
- User_ID → Name
- User_ID → Nationality
- User_ID → DOB
- User_ID → Password

BCNF Check:
Every FD has User_ID as its determinant. User_ID is the sole candidate key and therefore a superkey.
No partial dependencies (single-attribute PK means none are possible). No transitive dependencies — Name, Nationality, DOB, and Password are independent facts about a user.

USER is in BCNF.

2. USER_PHONE

Attributes: User_ID, Phone_no
Candidate Key(s):
{User_ID, Phone_no} — composite PK
All FDs: only key attributes present
BCNF Check:
A relation with only key attributes has no non-trivial FDs, so BCNF is trivially satisfied.

USER_PHONE is in BCNF.

3. USER_EMAIL

Attributes: User_ID, Email
Candidate Key(s):
{User_ID, Email} — composite PK
All FDs: only key attributes present
BCNF Check:
A relation with only key attributes has no non-trivial FDs, so BCNF is trivially satisfied.

USER_EMAIL is in BCNF.

4. JUDGE

Attributes: Judge_ID, User_ID, Working_status, Practising_since, Working_here_since

Candidate Key(s):

{Judge_ID} and {User_ID} — both are candidate keys.

Judge_ID is the primary key. User_ID is UNIQUE

All FDs:

- Judge_ID → User_ID
- Judge_ID → Working_status
- Judge_ID → Practising_since
- Judge_ID → Working_here_since
- User_ID → Judge_ID

No non-key attribute determines another non-key attribute.

JUDGE is in BCNF.

5. JUDGE_SPECIALISATION

Attributes: Judge_ID, Specialisation

Candidate Key(s):

{Judge_ID, Specialisation} — composite PK

All FDs: only key attributes present

BCNF Check:

No non-trivial FDs exist. BCNF is trivially satisfied.

JUDGE_SPECIALISATION is in BCNF.

6. LAWYER

Attributes: Bar_council_number, User_ID, Firm, Working_status

Candidate Key(s):

{Bar_council_number} and {User_ID} — both are candidate keys.

Bar_council_number is the primary key. User_ID is UNIQUE

All FDs:

- Bar_council_number → User_ID
- Bar_council_number → Firm
- Bar_council_number → Working_status
- User_ID → Bar_council_number

No non-key attribute determines another non-key attribute.

LAWYER is in BCNF.

7. LAWYER_SPECIALISATION

Attributes: Bar_council_number, Specialisation
Candidate Key(s):
{Bar_council_number, Specialisation} — composite PK
All FDs: only key attributes present
BCNF Check:
No non-trivial FDs exist. BCNF is trivially satisfied.

LAWYER_SPECIALISATION is in BCNF.

8. ADMIN

Attributes: Admin_ID, User_ID, Post, Working_here_since
Candidate Key(s):
{Admin_ID} and {User_ID} — both are candidate keys.
Admin_ID is the primary key. User_ID is UNIQUE
All FDs:

- Admin_ID → User_ID
- Admin_ID → Post
- Admin_ID → Working_here_since
- User_ID → Admin_ID (User_ID is UNIQUE → determines the whole tuple)

BCNF Check:
No non-key attribute determines another non-key attribute.

ADMIN is in BCNF.

9. LITIGANT

Attributes: User_ID, Litigant_ID, Government_ID, Type, ID_number
Candidate Key(s):
{User_ID} and {Litigant_ID} — both are candidate keys.
All FDs:

- User_ID → Litigant_ID
- Litigant_ID → Government_ID
- Litigant_ID → Type
- Litigant_ID → ID_number

BCNF Check:
No non-key attribute determines another non-key attribute.

LITIGANT is in BCNF.

10. CASE

Attributes: Case_ID, CNR, Case_title, Case_type, Filing_date, Current_status, Description

Candidate Key(s):

{Case_ID} — sole candidate key. CNR is a plain attribute with a unique value (not a candidate key).

All FDs:

- Case_ID → CNR
- Case_ID → Case_title
- Case_ID → Case_type
- Case_ID → Filing_date
- Case_ID → Current_status
- Case_ID → Description

BCNF Check:

No non-key attribute determines another non-key attribute.

CASE is in BCNF.

11. CAUSE_LIST

Attributes: List_ID, Courtroom_no, Date, Time

Candidate Key(s):

{List_ID}

All FDs:

- List_ID → Courtroom_no
- List_ID → Date
- List_ID → Time

BCNF Check:

No non-key attribute determines another.

CAUSE_LIST is in BCNF.

12. CASE_HISTORY (Weak Entity)

Attributes: Case_ID, Court_ID, Date, Verdict, Court_type

Candidate Key(s):

{Case_ID, Court_ID, Date} — composite PK.

Case_ID is the owner FK; Court_ID and Date are the discriminators.

All FDs:

- (Case_ID, Court_ID, Date) → Verdict
- (Case_ID, Court_ID, Date) → Court_type

BCNF Check:

No transitive dependencies within this relation.

CASE_HISTORY is in BCNF.

13. INTERLOCUTORY_APPLICATIONS

Attributes: IA_ID, Case_ID, Application_type, Status, Date
Candidate Key(s):
{IA_ID}
All FDs:
- IA_ID → Case_ID
- IA_ID → Application_type
- IA_ID → Status
- IA_ID → Date

BCNF Check:
All FDs have IA_ID as determinant, which is the sole candidate key.

INTERLOCUTORY_APPLICATIONS is in BCNF.

14. HEARING

Attributes: Hearing_ID, Case_ID, Next_date, Judge_ID, Verdict, Prosecutor_ID, Purpose, Defense_lawyer_ID, List_ID
Candidate Key(s):
{Hearing_ID}
All FDs:
- Hearing_ID → Case_ID
- Hearing_ID → Next_date
- Hearing_ID → Judge_ID
- Hearing_ID → Verdict
- Hearing_ID → Prosecutor_ID
- Hearing_ID → Purpose
- Hearing_ID → Defense_lawyer_ID
- Hearing_ID → List_ID

BCNF Check:
No transitive dependencies within this relation.

HEARING is in BCNF.

15. EVIDENCE

Attributes: Evidence_ID, Case_ID, Type, Submitted_by, Uploaded_by, Address, Submit_date
Candidate Key(s):
{Evidence_ID}
All FDs:
- Evidence_ID → Case_ID
- Evidence_ID → Type
- Evidence_ID → Submitted_by
- Evidence_ID → Uploaded_by

• Evidence_ID → Address
• Evidence_ID → Submit_date

BCNF Check:

All FDs have Evidence_ID as determinant, which is the sole candidate key.

EVIDENCE is in BCNF.

16. WITNESS

Attributes: Witness_ID, Name, Address, Gender, Witness_type

Candidate Key(s):

{Witness_ID}

All FDs:

• Witness_ID → Name
• Witness_ID → Address
• Witness_ID → Gender
• Witness_ID → Witness_type

BCNF Check:

All FDs have Witness_ID as determinant, which is the sole candidate key.

WITNESS is in BCNF.

17. WITNESS_PHONE

Attributes: Witness_ID, Phone_no

Candidate Key(s):

{Witness_ID, Phone_no} — composite PK

All FDs: only key attributes present

BCNF Check:

No non-trivial FDs exist.

WITNESS_PHONE is in BCNF.

18. WITNESS_EMAIL

Attributes: Witness_ID, Email

Candidate Key(s):

{Witness_ID, Email} — composite PK

All FDs: only key attributes present

BCNF Check:

No non-trivial FDs exist.

WITNESS_EMAIL is in BCNF.

19. PAYMENT (Weak Entity)

Attributes: Case_ID, Payment_no, Payer_name, Amount, Method, Date, Status, Purpose

Candidate Key(s):

{Case_ID, Payment_no} — composite PK.

Case_ID is the owner FK;

All FDs:

- (Case_ID, Payment_no) → Payer_name
- (Case_ID, Payment_no) → Amount
- (Case_ID, Payment_no) → Method
- (Case_ID, Payment_no) → Date
- (Case_ID, Payment_no) → Status
- (Case_ID, Payment_no) → Purpose

BCNF Check:

All FDs have the full composite PK {Case_ID, Payment_no} as determinant, which is the sole candidate key.

PAYMENT is in BCNF.

20. E_FILING

Attributes: Diary_number, Document_type, Petitioner_ID, Submit_date, Status, Document_title, Petitioner_lawyer_ID, Admin_ID, Case_ID

Candidate Key(s):

{Diary_number} and {Case_ID} — both are candidate keys.

Diary_number is the primary key. Case_ID has a UNIQUE constraint

All FDs:

- Diary_number → Document_type
- Diary_number → Petitioner_ID
- Diary_number → Submit_date
- Diary_number → Status
- Diary_number → Document_title
- Diary_number → Petitioner_lawyer_ID
- Diary_number → Admin_ID
- Diary_number → Case_ID
- Case_ID → Diary_number

BCNF Check:

No non-key attribute determines another non-key attribute.

E_FILING is in BCNF.

21. CASE_DOCUMENT

Attributes: document_id, case_id, document_type, filed_by, filed_date, hearing_id, document_address

Candidate Key(s):

{document_id}

All FDs:

- document_id → case_id
- document_id → document_type
- document_id → filed_by
- document_id → filed_date
- document_id → hearing_id
- document_id → document_address

BCNF Check:

All FDs have document_id as determinant, which is the sole candidate key.

CASE_DOCUMENT is in BCNF.

22. FIR

Attributes: FIR_number, police_station, district, date_of_filing, sections_invoked, accused_name, complainant_name, case_id

Candidate Key(s):

{FIR_number}

All FDs:

- FIR_number → police_station
- FIR_number → district
- FIR_number → date_of_filing
- FIR_number → sections_invoked
- FIR_number → accused_name
- FIR_number → complainant_name
- FIR_number → case_id

BCNF Check:

All FDs have FIR_number as determinant, which is the sole candidate key.

FIR is in BCNF.

23. HAS_WITNESS

Attributes: Case_ID, Witness_ID

Candidate Key(s):

{Case_ID, Witness_ID} — composite PK

All FDs: only key attributes present

BCNF Check:

No non-trivial FDs exist.

HAS_WITNESS is in BCNF.

24. <u>CASE_JUDGES</u>

Attributes: Case_ID, Judge_ID

Candidate Key(s):

{Case_ID, Judge_ID} — composite PK

All FDs: only key attributes present

BCNF Check:

No non-trivial FDs exist.

CASE_JUDGES is in BCNF.

25. <u>CASE_LITIGANTS</u>

Attributes: Case_ID, Litigant_ID

Candidate Key(s):

{Case_ID, Litigant_ID} — composite PK

All FDs: only key attributes present

BCNF Check:

No non-trivial FDs exist.

CASE_LITIGANTS is in BCNF.

26. <u>CASE_LAWYERS</u>

Attributes: Case_ID, Bar_council_number

Candidate Key(s):

{Case_ID, Bar_council_number} — composite PK

All FDs: only key attributes present

BCNF Check:

No non-trivial FDs exist.

CASE_LAWYERS is in BCNF.